

## Contact

hamnaraeel@gmail.com

[www.linkedin.com/in/hamnaraeel](https://www.linkedin.com/in/hamnaraeel)  
(LinkedIn)

## Top Skills

Data Science  
Microsoft Azure  
IBM Cloud

## Languages

English (Professional Working)  
Urdu (Native or Bilingual)

## Certifications

ChatGPT Prompt Engineering for Developers  
Frontend Fundamentals  
Developer Foundations With C++ Course  
Graphic Designer  
Certified AI, Metaverse and Web3.0 Developer & Solopreneur

# Hamna Raeel

Data Scientist | Research Assistant @AID-Lab | MS Data Science  
@FAST-NUCES | xIntern @SPS  
Islamabad, Islāmābād, Pakistan

## Summary

Hi, I'm Hamna Raeel, a Data Science graduate student (MS) and Research Assistant at the Artificial Intelligence Diagnostics Lab, FAST NUCES, passionate about leveraging Deep Learning, Machine Learning, and Computer Vision to solve real-world problems. I hold a Bachelor's degree in Computer Science and am currently pursuing my MS in Data Science, where I'm expanding my expertise in ML, DL, CV, and NLP. My ongoing research focuses on histopathology image classification and segmentation, utilizing models like Mask R-CNN, Fast R-CNN, ResNet, UNet, Vision Transformer (ViT), and Transformer-based architectures to enhance disease detection accuracy. Alongside research, I've gained professional experience in DevOps and Cloud Infrastructure, implementing CI/CD pipelines, Docker/Kubernetes orchestration, and cloud deployments across IBM Cloud, Azure, and VMware. I also have a strong foundation in Full-Stack Development, building scalable web applications using React, Next.js, Tailwind CSS, FastAPI, and TypeScript. Core Skills & Interests:- Deep Learning, Machine Learning & Computer Vision- Model Development, Evaluation & Deployment- Cloud & DevOps (Docker, Kubernetes, Jenkins, IBM Cloud)- Data Analysis, Visualization & Storytelling- Full-Stack Development (React, Next.js, TypeScript, FastAPI) I'm passionate about continuous learning and exploring how AI and data-driven technologies can advance healthcare, automation, and intelligent systems. Always open to collaboration, research, and data science opportunities that drive innovation and real-world impact.

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## Experience

Artificial Intelligence Diagnostics Lab FAST  
Research Assistant  
October 2025 - January 2026 (4 months)  
Islamabad

- Developed and trained deep learning segmentation models using Mask R-CNN, Fast R-CNN, SegFormer, and DETR.
- Optimized preprocessing, augmentation, and training pipelines to improve histopathology image segmentation performance.
- Built and evaluated deep learning classification models using ResNet, DenseNet, EfficientNet, Vision and Swin Transformer.
- Improved histopathology disease classification through feature extraction, model optimization, and performance evaluation.

## Software Productivity Strategists, Inc. (SPS)

Cloud Infrastructure & Containers, and DevOps & Automation Intern  
July 2025 - September 2025 (3 months)

Islāmābād, Pakistan

Developed and deployed a Python-based ETL pipeline for stock market data using the PyFinance API.

Built microservices architecture containerized with Docker and orchestrated using Kubernetes.

Automated deployment workflows using Jenkins CI/CD pipelines on IBM Cloud.

Worked with PostgreSQL, Prometheus, Grafana, and PaperTrail for database, monitoring, and logging.

Gained hands-on experience with VMware, Ubuntu Server, and Azure Cloud infrastructure.

## Artificial Intelligence Diagnostics Lab FAST

Data Science Intern

March 2025 - May 2025 (3 months)

Islāmābād, Pakistan

- Developed and evaluated ML/DL models using CNN, ResNet, UNet, SVM, and Decision Trees across benchmark datasets.
- Applied ML to medical imaging and retail datasets, including CBIS-DDSM, Chest X-Ray, Retinal Vessel, and CIFAR-10/100.
- Implemented RAG using vector databases, embedding models, and LLMs for intelligent document search and question answering.

## Optimus Bee Lahore

Software Engineer

November 2022 - December 2024 (2 years 2 months)

- Built an e-commerce platform with integrated Stripe payment processing, secure authentication, and order management.

- Developed a Mental Health Wellness application with an AI-powered chatbot, mood tracking, and recommendations.
- Created a Financial Budget Management Tool for expense tracking, categorization, and analytics dashboards.
- Built an admin dashboard with dynamic UI, REST APIs, user management, and reporting for an Australian travel agency.

## Cyber Legends

### Frontend Web Developer

June 2021 - October 2022 (1 year 5 months)

Lahore

As a Frontend Developer at Cyber Legends, I specialized in creating responsive, user-friendly digital platforms using WordPress, HTML, Tailwind CSS, and JavaScript. I led the development of a WordPress-based university portal, integrating custom functionalities like an Online Admission System and a responsive theme focused on aerospace and technology. This role involved managing detailed content for academic programs and campus life, implementing interactive study schemes, and creating a searchable faculty directory. Additionally, I developed the Infinite Writing freelance platform, constructing dual dashboards for clients and writers, and incorporating features such as an interactive price calculator and a dynamic review system and many other projects. My work focused on enhancing user experience and streamlining content management and administrative processes, significantly improving efficiency and user engagement across the platforms.

## Freelancer.com

### Freelance Graphic Designer

December 2019 - December 2020 (1 year 1 month)

Lahore District, Punjab, Pakistan

As a freelance graphic designer, I thrived on freelancer.com for a year, honing my skills in Adobe Illustrator, Adobe Photoshop, Adobe XD, and Figma. It was an enriching experience collaborating on diverse projects, pushing my creative boundaries, and delivering impactful designs. #FreelanceDesigner #CreativeJourney

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## Education

### National University of Computer and Emerging Sciences

Master of Science, Data Science · (August 2024)

Lahore College for Women University

Bachelor's degree, Computer Science · (September 2020 - September 2024)